



Radical Exponents and Square Roots

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Learning Objectives

- Convert between fractional exponents and radical notation
- Evaluate perfect square roots using the multiplication table
- Simplify irregular square roots using a factor tree
- Simplify radical expressions with variables

Simplify each expression completely, showing your work and writing answers in simplest radical form.

1. Evaluate the square root.

$$\sqrt{81}$$

Answer: _____

2. Evaluate the square root using the multiplication table.

$$\sqrt{144}$$

Answer: _____

3. Rewrite the fractional exponent as a radical and evaluate.

$$25^{\frac{1}{2}}$$

Answer: _____

4. Rewrite the fractional exponent as a radical and evaluate.

$$64^{\frac{1}{3}}$$

Answer: _____

5. Convert the radical to a fractional exponent.

$$\sqrt[4]{x^3}$$

Answer: _____

6. Simplify the radical using a factor tree.

$$\sqrt{72}$$

Answer: _____

7. Simplify the irregular square root.

$$\sqrt{50}$$

Answer: _____



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8. Simplify the irregular square root.

$$\sqrt{98}$$

Answer: _____

9. Simplify the radical expression with a variable.

$$\sqrt{75x^2}$$

Answer: _____

10. Simplify the radical expression completely.

$$\sqrt{200}$$

Answer: _____





This worksheet covers the full lesson on radicals: warm-up review of perfect squares from the multiplication table, converting fractional exponents to radicals, evaluating square roots of perfect squares, using a factor tree to simplify irregular square roots, and simplifying radical expressions with variable factors.

Solutions

1. Evaluate the square root.

$$\sqrt{81}$$

- Look for a number that multiplies by itself to give 81.
- Since 9 times 9 equals 81, the square root of 81 is 9.

Answer: 9

2. Evaluate the square root using the multiplication table.

$$\sqrt{144}$$

- Recall that 12 times 12 equals 144.
- Therefore the square root of 144 is 12.

Answer: 12

3. Rewrite the fractional exponent as a radical and evaluate.

$$25^{\frac{1}{2}}$$

- A fractional exponent of one-half means the square root.
- Rewrite as the square root of 25.
- Since 5 times 5 equals 25, the answer is 5.

Answer: 5

4. Rewrite the fractional exponent as a radical and evaluate.

$$64^{\frac{1}{3}}$$

- A fractional exponent of one-third means the cube root.
- Rewrite as the cube root of 64.
- Since 4 times 4 times 4 equals 64, the answer is 4.

Answer: 4

5. Convert the radical to a fractional exponent.

$$\sqrt[4]{x^3}$$

- The index of the root becomes the denominator of the fractional exponent.
- The exponent on the radicand becomes the numerator.
- Write the expression as x raised to the three-fourths power.

Answer: $x^{\frac{3}{4}}$



6. Simplify the radical using a factor tree.

$$\sqrt{72}$$

- Build a factor tree for 72 to get the prime factorization 2 times 2 times 2 times 3 times 3.
- Group the primes into pairs: one pair of 2s and one pair of 3s, with a single 2 left over.
- Each pair comes out of the radical as one factor, giving 2 times 3 outside and 2 inside.
- Multiply the outside numbers to get 6 times the square root of 2.

Answer: $6\sqrt{2}$

7. Simplify the irregular square root.

$$\sqrt{50}$$

- Factor 50 as 25 times 2, where 25 is a perfect square.
- Take the square root of 25 to get 5 outside the radical.
- The 2 remains inside, giving 5 times the square root of 2.

Answer: $5\sqrt{2}$

8. Simplify the irregular square root.

$$\sqrt{98}$$

- Factor 98 as 49 times 2, where 49 is a perfect square.
- Take the square root of 49 to get 7 outside the radical.
- The 2 remains inside, giving 7 times the square root of 2.

Answer: $7\sqrt{2}$

9. Simplify the radical expression with a variable.

$$\sqrt{75x^2}$$

- Factor 75 as 25 times 3, where 25 is a perfect square.
- The square root of 25 is 5 and the square root of x squared is x .
- Write $5x$ outside the radical with 3 remaining inside.

Answer: $5x\sqrt{3}$

10. Simplify the radical expression completely.

$$\sqrt{200}$$

- Use a factor tree to find the prime factorization of 200 as 2 times 2 times 2 times 5 times 5.
- Group the primes into pairs: one pair of 2s and one pair of 5s, with a single 2 left over.
- Each pair contributes one factor outside the radical, giving 2 times 5 outside and 2 inside.
- Multiply to get 10 times the square root of 2.

Answer: $10\sqrt{2}$

